The Oscilloscope

The oscilloscope is an electronic measuring instrument used to display the waveform of signal. Compared with other measuring instruments, oscilloscope has the following advantages: it can display the waveform of the measured signal, it has little effect on the system; it has a wide passband, high sensitivity and large dynamic range, strong overload capacity, etc. So, oscilloscope is widely used in industry, scientific research, national defense and other fields.

Objectives

1. Learn the working principle of oscilloscope.
2. Master the use of common functions of oscilloscope.
3. Learn the use of signal generator.

Apparatus

Oscilloscope, signal generator, unknown signal source.

Theory

1.The cathode ray tube (CRT)

The heart of the oscilloscope is the CRT. The general features of a cathode ray tube, which forms the heart of nearly every TV set, are shown schematically in Fig. 1 below. A beam of electrons traveling at an enormous speed is produced by a system consisting of an electron emitting filament (cathode) and a collection of focusing and accelerating electrodes, all collectively referred to as an electron gun. The widened end of the tube opposite the gun is coated with chemicals (phosphors) which glow when struck by the electron beam and produce a bright spot on the screen. The position of the spot is controlled by applying voltage to pairs of horizontal (1, 2) and vertical (3, 4) deflection plates. By simultaneously applying varying voltage to both pairs of plates, the position of the electron-beam spot can be made to trace out a curve on the face of the screen.

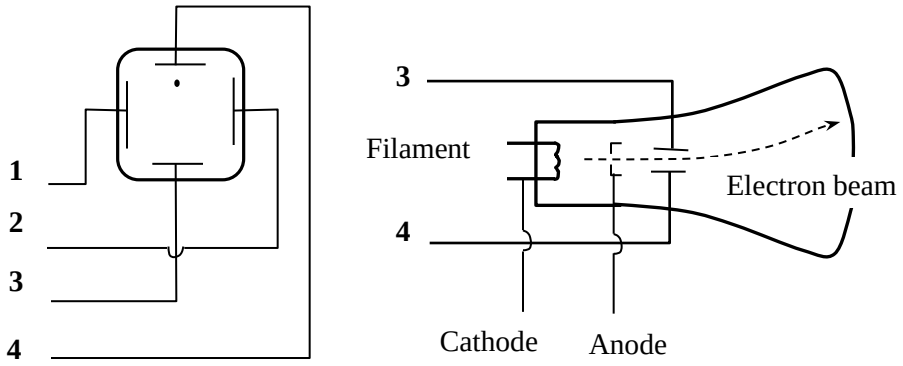

Fig. 1 The structure of CRT

2. Display of waveform

When a sinusoidal signal is added to the vertical deflection plate of the ray tube, the light spot will move along a vertical line on the screen, as shown in Fig. 2. At this time, if a voltage U(x), which is proportional to time, is added in the transverse direction, the motion track of the light spot will expand in the X direction, forming a waveform.

When the spot moves to the right end, it runs out of the display. In order to continue to display the waveform on the screen, U(x) should return to zero and start the second waveform expansion. Therefore, the waveform of U(x) signal should be similar to the sawtooth wave as shown in Fig. 3. U(x) voltage is called the sweep voltage.

When sawtooth voltage is applied to the horizontal deflector, only a bright horizontal line is displayed on the screen. If a sinusoidal voltage is applied to the vertical deflector plate and a sawtooth voltage is applied to the horizontal deflector plate, the electrons are acted on by both vertical and horizontal forces, and the motion of the electrons is the synthesis of the two perpendicular motions of each other. When the sawtooth voltage equals the period of the sinusoidal signal, a stable sinusoidal voltage waveform will be displayed on the screen, as shown in Fig. 4.

If a sinusoidal signal is applied to both vertical and horizontal deflector, a Lissajous pattern can be formed as long as the frequency ratio of the two signals is an integer, as shown in Fig. 5.

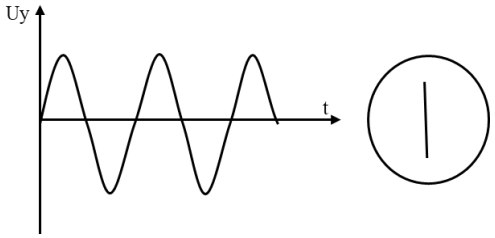

Fig. 2 A sinusoidal signal is added to the vertical deflection plate

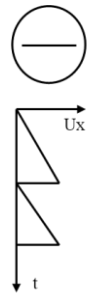

Fig. 3 Sawtooth wave

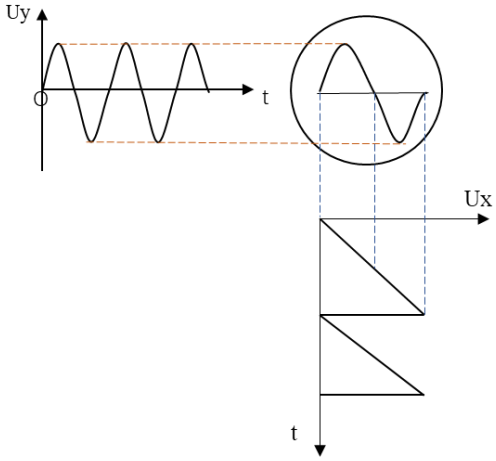

Fig. 4 Principle of sine wave

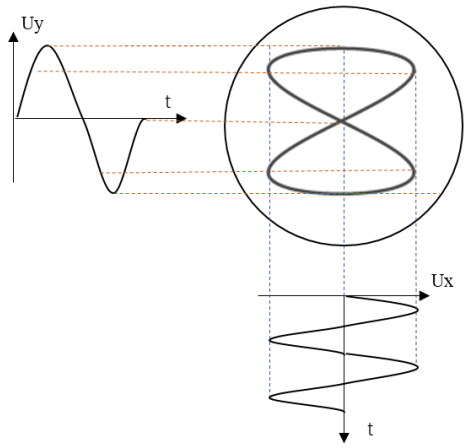

Fig. 5 Principle of Lissajous pattern

3. Synchronization of scanning voltage and trigger

To obtain a stable pattern on the CRT screen, the sawtooth wave has to be in phase with the signal that you want to observe. This is done with a trigger circuit. The trigger circuit allows the oscilloscope to draw repeatedly the same waveform over and over by identifying the same point on a repetitive waveform. A certain voltage is set as the trigger level, and when the voltage rises to the trigger level in each cycle (Fig. 6(a)), a sharp pulse is generated, which generates a pulse wave with the same frequency as the signal (Fig. 6 (b)). After the scanning voltage returns to the minimum value, it should not start until the pulse is triggered, as shown in Fig. 6 (c). This ensures that the scanning voltage is synchronized with the signal.

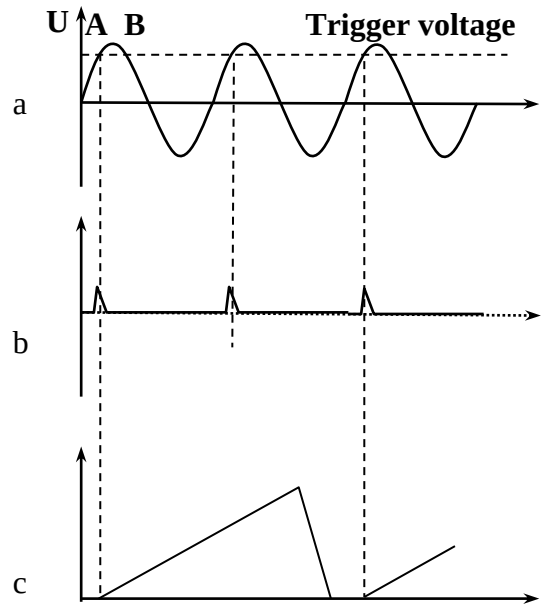

Fig. 6 The working of trigger

4. Trigger technique

A brief explanation of the trigger principle is given above. In fact, trigger source, trigger level, trigger polarity and trigger mode are varied. Trigger adjustment technique is one of the more difficult techniques for beginners to master when using oscilloscope.

(1) Trigger source

The trigger source includes internal trigger, external trigger, power trigger, etc. Internal trigger uses signal as trigger source. In the dual-channel oscilloscope, there are channel 1 (CH1) trigger, channel 2 (CH2) trigger and CH1, CH2 double trigger. If the signals of the two channels have a fixed phase relationship, stable waveforms can be obtained regardless of using CH1 or CH2 trigger. If the two channel signals are independent and have no fixed phase relationship, double trigger must be used to display two waveforms at the same time. The external trigger is made by the signal outside the oscilloscope as the trigger source, which is used when the internal trigger is not ideal.

(2) Trigger level, rising and falling edge and trigger polarity

The trigger level is adjustable. Sometimes it is necessary to adjust the trigger level with the amplitude and waveform of the signal to obtain the ideal trigger effect. The same trigger level also has rising edge trigger and falling edge trigger (point A and B in Fig. 6). According to the positive and negative trigger level, there are positive and negative polarity trigger. Different choice of trigger level up and down edge and polarity will change the starting point of the display waveform.

(3) Trigger mode

Trigger mode includes normal, automatic, and single trigger. In normal mode, it scans in the X direction only when the trigger signal exists, otherwise, there is no signal on the screen. In automatic mode, if trigger signals exist, it scans according to the preceding principles. If there is no trigger signal, the oscilloscope will scan automatically, and there is waveform display on the screen, but it is not stable. The single scan mode generates signal only once, which is mainly used to observe monopulse signal and aperiodic signal.

5. Front panel of oscilloscope

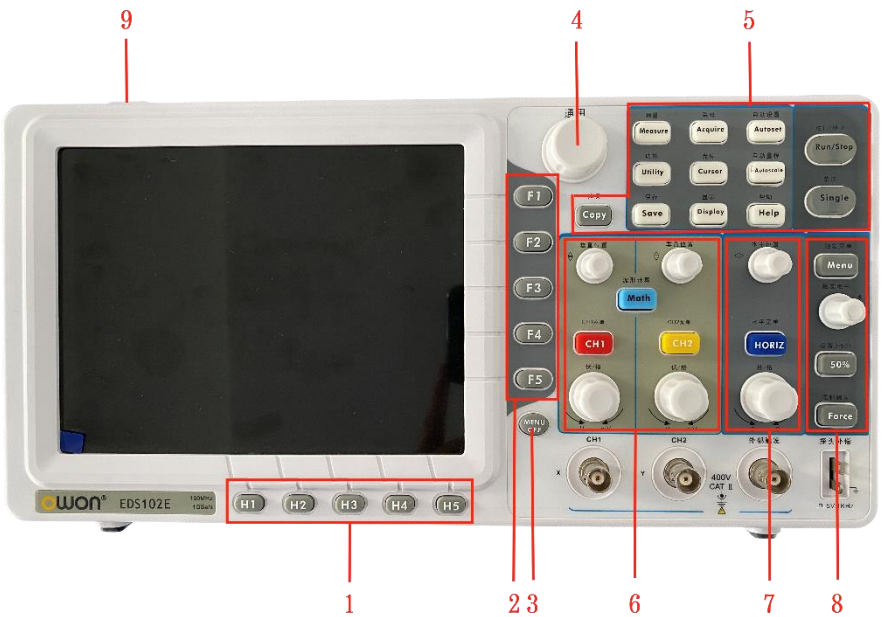

Fig. 7 Front panel of oscilloscope

1- The option area of the horizontal menu contains five buttons: H1 to H5; 2-The option area of the vertical menu contains five buttons: F1 to F5; 3-Menu off: Close the menu displayed on the screen; 4-Universal button: Turn the button to select the menu or set the value. Press the button to close the menu on the left side of the screen; 5- Function area; 6-Vertical control area: Two vertical position buttons control the vertical displacement of CH1 and CH2 respectively. Two V/DIV buttons control the voltage gears in CH1 and CH2 respectively; 7-Horizontal control area: The horizontal position button controls the horizontal position of the trigger, and the Time/DIV button controls the time base gear; 8-Trigger control area: Adjust the trigger level and set the trigger system; 9-ON/OFF.

The application of oscilloscope

1. Measure the voltage of the signal

Y is the deflection of the electron beam along the Y axis, which can be represented by the number of cells (div). k_y is the voltage deflection factor (V/div) of the Y axis. The voltage of the signal can be expressed as

$$U=Yk_y \tag{1}$$

The voltage measurement method is shown in Fig. 8.

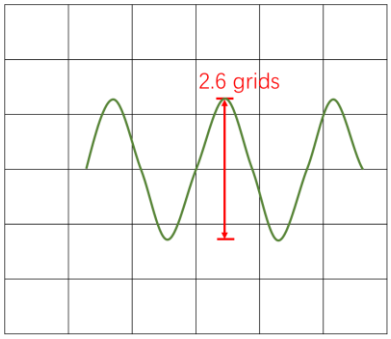

Fig. 8 Voltage measurement

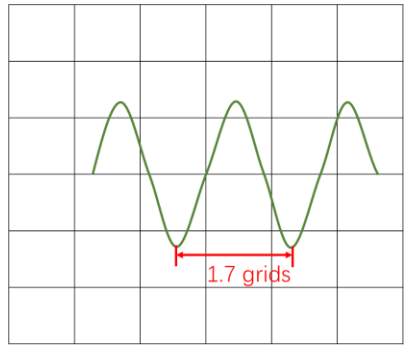

Fig. 9 Frequency measurement

2. Measure the frequency of the signal

(1) Period conversion method

Period and frequency are reciprocals of each other: $f = \frac{1}{T}$. When x_t is the length of a period on the screen, represented by the grids (div), and R_t is the scanning time factor (ms/div), then the frequency of the signal is $\mathrm{T} = x_t \cdot R_t$ (Fig. 9)

(2) Lissajous pattern method

The method of frequency comparison by Lissajous pattern is called Lissajous pattern method. This method was proved by Lissajous in 1855.

The unknown signal is fed into the vertical system and the standard signal into the horizontal system. When the frequency ratio of the two is an integer, a Lissajous pattern is displayed on the screen. If the standard frequency is known, the frequency of the unknown signal can be calculated from the pattern. How to know the ratio of two sinusoidal voltage frequencies? Make horizontal and vertical tangents when the Lissajous pattern on the screen is stable. The intersection number of horizontal and vertical lines is N_x and N_y respectively, then

$$\frac{f_y}{f_x} = \frac{N_x}{N_y} \tag{2}$$

Fig. 10 shows that the tangent points of the horizontal and vertical lines are 3 and 2 respectively, so $\frac{f_y}{f_x} = \frac{3}{2}$.

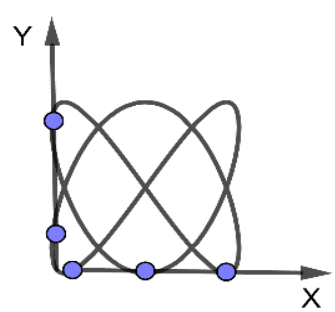

Fig. 10 The Lissajous pattern of 3:2 measurement

Contents

1. Draw sinusoidal waves, triangular waves and square waves of different frequencies respectively in three coordinate systems.

2. Measure the voltage of unknown signal with oscilloscope, record it in Table 1. The valid value retains three significant digits.

3. Measure the frequency of unknown signals, record it in Table 2. The frequency retains four significant digits.

4. The frequency of unknown signal is measured by Lissajous pattern method. The two input terminals of oscilloscope are connected with sinusoidal signals, and Lissajous patterns can be obtained in X-Y mode. Change the standard signal and observe the change of the pattern. The frequency of the unknown signal is measured when the frequency ratio of the standard signal and the unknown signal is 1:1, 1:2, 2:1 and 3:2, respectively. Record the frequency of standard signal, Lissajous pattern, and calculate the unknown signal frequency value in Table 3.

Precautions

1. The voltage and time gear do not need to be evaluated when reading, and the grids of waveform need to be evaluated according to the minimum scale.

2. Pay attention to the adjustment of trigger source, trigger mode and trigger level.

3. When measure the frequency with Lissajous pattern method, record the frequency after the Lissajous pattern is still.

Data Taking

Table 1 Voltage measurement

VOLT/DIV	Grids	Peak-Peak Value（V）	Effective Value（V）
___V/DIV			
___V/DIV			
___V/DIV			

Table 2 Frequency measurement

TIME/DIV	Grids	Period（ms）	Frequency（Hz）
___ms/DIV			
___ms/DIV			
___ms/DIV			

Table 3 Frequency measurement with Lissajous pattern method

Standard frequency : Unknown frequency (f_x:f_y)	Standard frequency f_x (Hz)	Lissajous pattern	Unknown frequency f_y(Hz)
1:1			
1:2			
2:1			
3:2			

Questions

1. When is Lissajous pattern a circle?
2. Where should be checked if no image on the oscilloscope screen?